



AE703: Computational Methods
Final-exam programming part
Due Date: 16/11/25

1. Solve the equation $\frac{dy}{dt} = -y + e^t$, $y(0) = 1$ using 4th order Adam-Bashforth method with $h=0.1$ & 0.2 for t in $(0,1)$.

[10 points]

2. Solve the following Poisson Boltzman equation using the shooting method with 4th order Runge Kutta method

$$\frac{d^2\Psi}{dx^2} = \sinh \Psi, \Psi(0) = 1, \Psi(x) \rightarrow 0 \quad \text{and} \quad \frac{d\Psi}{dx} \rightarrow 0 \text{ as } x \rightarrow 0 \quad .$$

[10 points]



AE703: Computational Methods
Quiz (Date: 03/11/2025)
Duration: 15 minutes

1. Solve the equation $y' = y - x^2 + 1$, $y(0) = 2$ using 2-step Adam-Bashforth method with $h=0.1$. Find out y_2 .

[10 points]



AE703: Computational Methods
Quiz (Date: 03/11/2025)
Duration: 15 minutes

1. Solve the equation $y' = -x^2 + 1 + y$, $y(0) = 1.5$ using 2-step Adam-Bashforth method with $h=0.1$. Find out y_2 .

[10 points]



AE703: Computational Methods
Quiz (Date: 03/11/2025)
Duration: 15 minutes

1. Solve the equation $y' = -x^2 + 1 + y$, $y(0) = 2$ using 2-step Adam-Bashforth method with $h=0.2$. Find out y_2 .

[10 points]



AE703: Computational Methods
Quiz (Date: 03/11/2025)
Duration: 15 minutes

1. Solve the equation $y' = y - x^2 + 1$, $y(0) = 2$ using 2-step Adam-Bashforth method with $h=0.15$. Find out y_2 .

[10 points]

